

MODE OF TRANSACTION

- Lecture, Seminar, Discussion, Questioning and Answering

ASSESSMENT RUBRICS

Refer to section 7 of FYIMP- Computational Science - Scheme and Syllabus for the 4 credit courses (4 Credit theory).

Sample Questions to test Outcomes.

1. How do you handle software requirements changes during a project?
2. Can you describe the Agile software development methodology and its advantages?
3. What is the difference between unit testing and integration testing?
4. Explain concept of data flow diagram.
5. Give a description of the prototyping model.

KU5DSCCSE305 MACHINE LEARNING TECHNIQUES

Semester	Course Type	Course Level	Course Code	Credits	Total Hours
5	DSC	300	KU5DSCCSE305	4	90

Learning Approach (Hours/ Week)			Marks Distribution			Duration of ESE (Hours)
Lecture	Practical/ Internship	Tutorial	CE	ESE	Total	

2	4	1	50	50	100	2(T)+3(P)*
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*** ESE duration : 2 hours for theory and 3 hours for Lab**

Course Description:

This course is intended to familiarize the learners about the fundamental concepts in machine learning and deep learning mechanism for applying and analysing the huge volume of data generated in real life. The main purpose of this course is to provide a foundational understanding of machine learning models and demonstrate how it will be useful for solving complex problems in the real world.

Course Objectives:

- Aims to impart basic concepts of Artificial intelligence
- Impart knowledge on different learning scenario in machine learning
- Acquire the knowledge of various machine learning models and its implementation
- Understand various neural networks models architecture and its training algorithms
- Awareness of various dimensionality reduction techniques for optimizing the feature extraction techniques for improving the performance of various classification models.

Course Outcomes:

At the end of the Course, the Student will be able to:

SL #	Course Outcomes
CO1	To introduce the prominent methods for machine learning
CO2	To study the basics of supervised and unsupervised learning

CO3	To study the basics of neural networks
CO4	To study the basics of deep learning architectures

Mapping of COs to PSOs

	PSO1	PSO2	PSO3	PSO4	PSO5
CO1	✓	✓	✓	✓	✓
CO2	✓	✓	✓	✓	✓
CO3	✓	✓	✓	✓	✓
CO4	✓	✓	✓	✓	✓

COURSE CONTENTS

Module 1: Introduction to Machine Learning: Concept of the learning task, inductive learning, and the concepts of hypothesis space, introduction to different types of machine learning approaches, examples of machine learning applications, different types of learning: supervised learning, unsupervised learning, reinforcement learning. Feature Extraction- Setting up your machine learning platform: training, validation and testing, overfitting and under-fitting, different types of error calculation. (15 hours)

Module 2: Regression- Linear regression and logistic regression- Supervised Learning: Introduction, learning a class from example, learning multiple classes, Unsupervised Learning: Introduction, clustering; mixture densities, k-means clustering and hierarchical clustering. (20 hours)

Module 3: Dimensionality reduction: principal component analysis, linear discriminant analysis, canonical correlation analysis. Introduction to Artificial Neural Network: Understanding brain, perceptron, Multi-Layer perceptron as

universal approximator, general architecture of artificial neural network, feed forward and back propagation, different linear and nonlinear activation functions for binary and multi class classification. (25 hours)

Module 4: Introduction to Deep Learning: Fundamentals of deep learning, Deep Feedforward Networks, Regularization for Deep Learning, Optimization for Training Deep Models, Introduction to Convolutional Networks, Sequence Modelling using Recurrent Nets, overview of LSTM, fundamentals of Generative adversarial Networks. (25 hours)

Module X: Feature Extraction - Testing the quality of features Reinforcement Learning, Transfer Learning Anomaly Detection and Outlier Analysis, Natural Language Processing (NLP), Graph Neural Networks (GNNs), Adversarial Machine Learning

Core Compulsory Readings

1. Ethem Alpaydin, Introduction to Machine Learning- 3rd Edition, PHI.
2. Ian Goodfellow and Yoshua Bengio and Aaron Courville, Deep Learning (Adaptive Computation and Machine Learning), MIT Press, 2016.

Core Suggested Readings

1. Tom M. Mitchell, Machine Learning, McGraw-Hill
2. Kuntal Ganguly, Learning Generative Adversarial Networks, Packt Publishing, 2017.

TEACHING LEARNING STRATEGIES

- Lecturing, Demonstration, Digital Learning, Team Work

MODE OF TRANSACTION

- Lecture, Seminar, Discussion

ASSESSMENT RUBRICS

Refer to section 7 of FYIMP- Computational Science - Scheme and Syllabus for the 4 credit courses with 2 Credit Theory + 2 Credit Practical.

Sample Questions to test Outcomes

1. What are the key differences between supervised, unsupervised, and reinforcement learning? Provide examples of real-world applications for each type of learning.
2. Explain the concepts of over-fitting and under-fitting in machine learning models. How can these issues be addressed during the training and validation process?
3. Compare and contrast linear regression and decision tree learning approaches in supervised learning. When would you choose one over the other for a given problem?
4. How does model selection impact the generalization performance of a supervised learning algorithm? Discuss techniques for selecting the best model among multiple candidates.
5. Describe the process of clustering using the k-means algorithm. What are the advantages and limitations of k-means clustering compared to hierarchical clustering?
6. Explain how Latent Dirichlet Allocation (LDA) is used for topic modeling in unsupervised learning. What are some real-world applications of LDA?
7. What are the primary objectives of dimensionality reduction techniques such as principal component analysis (PCA) and linear discriminant analysis (LDA)? How do these techniques differ in their approach?
8. Discuss the architecture and training process of a Multi-Layer Perceptron (MLP) neural network. How do feedforward and backpropagation algorithms contribute to learning in MLPs?